Simplified approach to extracting nucleon transversity in collinear factorization using near-side energy-energy correlators

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We develop a novel strategy for accessing the transversity parton distribution function (PDF) of the nucleon within collinear factorization using near-side energy-energy correlators in the dihadron fragmentation framework. We show how this removes the complications of previous approaches that must model either intrinsic parton transverse momentum or resonances in the invariant mass distribution of a final-state dihadron. We present leading-order analytical results for transverse-spin observables in semi-inclusive deep-inelastic scattering and electron-positron annihilation, highlighting their close similarity to the expressions one uses in extracting (un)polarized PDFs and (single-hadron) fragmentation functions in collinear factorization. We make predictions for Electron-Ion Collider, BELLE, and BESIII kinematics that demonstrate the feasibility of an energy-energy correlator program in extracting the transversity PDF.

Introduction and Motivation—

EEC Dihadron Fragmentation Functions— We start with the operator definitions [1] (see also Ref. [2]) of the two DiFFs, $D_1^{h_1h_2/q}$ and $H_1^{\sphericalangle\, h_1h_2/q}$, that will be relevant for our analytical results. Using the correlator

$$\Delta^{h_1h_2\,[\Gamma]} \equiv \sum_X\!\!\!\!\!\!\int \frac{dx^+ d^2\vec{x}_\perp}{(2\pi)^3} e^{ik\cdot x} \times \mathrm{Tr}\langle 0|\Gamma\,\psi(x)|P_1,P_2;X\rangle\langle P_1,P_2;X|\bar{\psi}(0)|0\rangle\big|_{x^-=0}\,, \quad (1)$$

we have

$$D_1^{h_1h_2/q}(\xi_1,\xi_2,\vec{R}_T) = \frac{\xi^2}{64\pi^3\xi_1\xi_2}\int d^2\vec{k}_T\,\Delta^{h_1h_2\,[\gamma^-]}\,, \quad (2)$$

$$-\frac{\epsilon_T^{ij}R_T^j}{M_h}H_1^{\sphericalangle\, h_1h_2/q}(\xi_1,\xi_2,\vec{R}_T) = \frac{\xi^2}{64\pi^3\xi_1\xi_2}\int d^2\vec{k}_T\,\Delta^{h_1h_2\,[i\sigma^{i-}\gamma^5]}\,, \quad (3)$$

where ξ_1, ξ_2 are the fractions carried by the hadrons h_1, h_2 of the lightcone momentum of the fragmenting quark q (with momentum k), and $\vec{R}_T \equiv \frac{1}{2}(\vec{P}_{1T} - \vec{P}_{2T})$ is (half of) their relative transverse momentum. (We have suppressed gauge links for brevity and a color average is understood.) The total momentum of the dihadron is denoted by P_h, with $M_h^2 = P_h^2$ its invariant mass, and $\xi \equiv \xi_1 + \xi_2$. The Levi-Civita tensor is defined as $\epsilon_T^{ij} \equiv \epsilon^{-+ij}$, with $\epsilon^{0123} = +1$. We work in the "dihadron frame" where P_h has no transverse momentum and has a large lightcone-minus component.

In Ref. [3] (see also Refs. [CITE]) we introduced the "EEC-DiFF" $\mathcal{D}^i(z_\chi, Q^2)$ $(i = q, g)$, where $z_\chi \equiv \frac{1}{2}(1 - \cos\chi)$ and Q is the relevant hard scale, that was able to well describe near-side EECs in e^+e^- annihilation. In calculating transverse-spin EEC observables involving dihadron final states, due to the fact that $H_1^{\sphericalangle\, h_1h_2/q} = -H_1^{\sphericalangle\, h_2h_1/q}$ [CITE], it becomes necessary to specify the species of the two hadrons being detected, rather than summing over h_1, h_2. Therefore, we define the following non-perturbative functions for our analysis:

$$\mathcal{D}^{h_1h_2/i}(z_\chi,Q^2) \equiv \int d\xi_1\int d\xi_2\int d^2\vec{R}_T\,\delta\!\left(z_\chi - \frac{R_T^2}{Q^2}\frac{\xi^2}{\xi_1^2\xi_2^2}\right) \times\, \xi_1\xi_2\,D_1^{h_1h_2/i}(\xi_1,\xi_2,\vec{R}_T)\,, \quad (4)$$

$$\mathcal{H}^{h_1h_2/i}(z_\chi,Q^2) \equiv \int d\xi_1\int d\xi_2\int d^2\vec{R}_T\,\delta\!\left(z_\chi - \frac{R_T^2}{Q^2}\frac{\xi^2}{\xi_1^2\xi_2^2}\right) \times\, \xi_1\xi_2\,\frac{R_T}{M_h}H_1^{\sphericalangle\, h_1h_2/i}(\xi_1,\xi_2,\vec{R}_T)\,, \quad (5)$$

where $R_T \equiv |\vec{R}_T|$. The evolution equation for $\mathcal{D}^{h_1h_2/i}(z_\chi,Q^2)$ was derived in Ref. [3]. The one for $\mathcal{H}^{h_1h_2/i}(z_\chi,Q^2)$ takes the same form, but the transversely polarized splitting kernels [4] are used instead [1, 5].

Transverse-Spin Observables for Near-Side EECs— We will study transverse-spin dependent near-side EECs for SIDIS, $e(l)\,N^\uparrow(P) \to e(l')\,(h_1(P_1)\,h_2(P_2))\,X$, and e^+e^- annihilation to two almost back-to-back dihadrons, $e^+(l')e^-(l) \to (h_1(P_1)\,h_2(P_2))\,(\bar{h}_1(\bar{P}_1)\bar{h}_2(\bar{P}_2))\,X$, where the momenta of the particles involved are given in parentheses. For both processes we introduce the variable $z_{12} \equiv \frac{1}{2}\Big(1 - \frac{\vec{P}_1\cdot\vec{P}_2}{|\vec{P}_1||\vec{P}_2|}\Big) = \frac{1}{2}(1-\cos\theta_{12})$, where θ_{12} is the angle between h_1 and h_2. Up to power suppressed (p.s.) corrections $\sim 1/Q^2$, one can show,

$$z_{12} = \frac{R_T^2}{Q^2}\frac{\tau^2}{\tau_1^2\tau_2^2}\,, \quad (6)$$

with $\tau \equiv \tau_1 + \tau_2$. For SIDIS, $\tau_{1(2)} = P\cdot P_{1(2)}/P\cdot q$, with $q = l - l'$ and $Q^2 = -q^2$, whereas for e^+e^- annihilation, $\tau_{1(2)} = 2P_{1(2)}\cdot q/Q^2 = 2E_{1(2)}/Q$, where $E_{1(2)}$ is the energy of hadron $h_{1(2)}$, and $Q^2 = q^2$ is the squared center-of-mass (c.m.) energy of the collision.

We define the transverse-spin EEC for SIDIS as

$$\mathrm{EEC}_{\mathrm{SIDIS}}^{h_1h_2} \equiv \int d\tau_1 d\tau_2 d^2\vec{R}_T\, \tau_1\tau_2 \times \frac{d\sigma^{\mathrm{SIDIS}}}{dxdyd\tau_1 d\tau_2 d^2\vec{R}_T d\phi_S}\, \delta(z_\chi - z_{12})\delta(\phi-\phi_{R_T})\,, \quad (7)$$

where $x = Q^2/2P\cdot q$ and $y = P\cdot q/P\cdot l$, and for e^+e^- annihilation as

$$\mathrm{EEC}_{e^+e^-}^{h_1h_2,\bar{h}_1\bar{h}_2} \equiv \int d\tau_1 d\tau_2 d^2\vec{R}_T \int d\bar{\tau}_1 d\bar{\tau}_2 d^2\vec{\bar{R}}_T\, (\tau_1\tau_2)(\bar{\tau}_1\bar{\tau}_2) \times \frac{d\sigma^{e^+e^-}}{d\tau_1 d\tau_2 d^2\vec{R}_T d\bar{\tau}_1 d\bar{\tau}_2 d^2\vec{\bar{R}}_T dy} \times \delta(z_\chi - z_{12})\delta(z_{\bar{\chi}} - z_{\overline{12}})\delta(\phi-\phi_{R_T})\delta(\bar{\phi}-\phi_{\bar{R}_T})\,, \quad (8)$$

where $y = P_h\cdot l/P_h\cdot q$. The leading-order results for the near-side dihadron contribution to $\mathrm{EEC}_{\mathrm{SIDIS}}^{h_1h_2}$ and $\mathrm{EEC}_{e^+e^-}^{h_1h_2,\bar{h}_1\bar{h}_2}$ can be found by following calculations of transverse-spin dependent dihadron cross sections [CITE], although in this case one is differential in $(\tau_1, \tau_1, \vec{R}_T)$ instead of (τ, M_h). The final expressions read, respectively,

$$\mathrm{EEC}_{\mathrm{SIDIS}}^{h_1h_2} \overset{\mathrm{DiFF}}{=} \frac{\alpha_{em}^2}{\pi y Q^2}\sum_{q,\bar{q}} e_q^2 \left[(1-y+\tfrac{y^2}{2}) f_1^{q/N}(x)\,\mathcal{D}^{h_1h_2/q}(Z) - (1-y)\sin(\phi+\phi_S) h_1^{q/N}(x)\mathcal{H}^{h_1h_2/q}(Z)\right] \quad (9)$$

$$\mathrm{EEC}_{e^+e^-}^{h_1h_2,\bar{h}_1\bar{h}_2} \overset{\mathrm{DiFF}}{=} \frac{N_c\alpha_{em}^2}{\pi Q^2}\sum_{q,\bar{q}} e_q^2 \left[(y^2-y+\tfrac{1}{2})\mathcal{D}^{h_1h_2/q}(Z)\mathcal{D}^{\bar{h}_1\bar{h}_2/\bar{q}}(\bar{Z}) + y(1-y)\cos(\phi+\bar{\phi})\mathcal{H}^{h_1h_2/q}(Z)\mathcal{H}^{\bar{h}_1\bar{h}_2/\bar{q}}(\bar{Z})\right], \quad (10)$$

where we have introduced the shorthand $Z \equiv z_\chi Q^2$ to emphasize that, due to the delta functions and integrations over $(\xi_1, \xi_2, \vec{R}_T)$ in Eqs. (4), (5), $\mathcal{D}^{h_1,h_2/i}$ and $\mathcal{H}^{h_1,h_2/i}$ only depend on the *single variable* $z_\chi Q^2$ (up to an overall prefactor that depends just on Q^2). The structure of Eqs. (9), (10) is exactly the same as *collinear factorization* in SIDIS and e^+e^- for *single-hadron* observables that are widely used in extractions of (un)polarized PDFs and FFs [CITE], where one encounters convolutions like $f_1(x)\otimes D_1(z)$ or $g_1(x)\otimes D_1(z)$. We note also that the unpolarized result up to next-to-leading order for the DiFF contribution to the near-side EEC in $e^+e^- \to h_1h_2\,X$ was given in Ref. [3]. For the SIDIS case we present the next-to-leading order analytical expression in Supplemental Material as well as provide predictions for the EIC based on the EEC-DiFF $\mathcal{D}^i(z_\chi, Q)$ extracted in Ref. [3].

The form of Eqs. (9), (10) highlights the advantage of our proposed program to extract the transversity PDF through those observables. Currently, a widely-used procedure relies on transverse-spin processes described by transverse momentum dependent (TMD) factorization with single-hadron final states [CITE]. These reactions involve the transversity TMD $h_1(x, k_T)$, unpolarized TMD $f_1(x, k_T)$, Collins TMD FF $H_1^\perp(z, p_\perp)$, and the unpolarized TMD FF $D_1(z, p_\perp)$, whose dependence on the intrinsic transverse momentum k_T and $p_\perp$ must be modeled, oftentimes through complicated functional forms in the Fourier-conjugate "b-space" [CITE]. This statement also applies to the proposed extraction of $h_1(x)$ using one-point energy correlators for hadron-in-jet observables in proton-proton collisions [CITE].

The other common approach to extracting the transversity PDF is through transverse-spin observables with dihadron final states that can be analyzed in collinear factorization [CITE]. In this case, the collinear PDFs $h_1(x)$ and $f_1(x)$ enter along with the DiFFs $H_1^\sphericalangle(z, M_h)$ and $D_1(z, M_h)$. This strategy was initially conceived as a way to bypass the complications that arise from TMD factorization [CITE]. Nevertheless, one still must confront the complicated M_h dependence of $H_1^\sphericalangle(z, M_h)$ and $D_1(z, M_h)$ due to resonance channels that arise, which requires $O(100)$ parameters to model [CITE]. On the other hand, the EEC-DiFFs in Eqs. (9), (10) have no dependence on M_h. Based on our work in Ref. [3], we anticipate $\mathcal{D}^{h_1h_2/i}(Z)$ and $\mathcal{H}^{h_1h_2/i}(Z)$ will each take on a simple functional form $\sim NQ^2\exp(-Z/\alpha)/(1+Z/\beta)$ (N, α, β being free parameters that possibly depend on flavor), making the phenomenological procedure for extracting $h_1(x)$ use EECs essentially the same as that for $f_1(x)$ and $g_1(x)$.

[AM: We also have A_{TT} for Drell-Yan and other final states in pp. Furthermore, there is polarized Λ production in SIDIS, and also pp. (Honestly, I had forgotten about these.) Because of that we may have to stress more the point of a "realistic and simple" observable. People could use existing data and reanalyze them. This may not be the case for any of the other observables that are

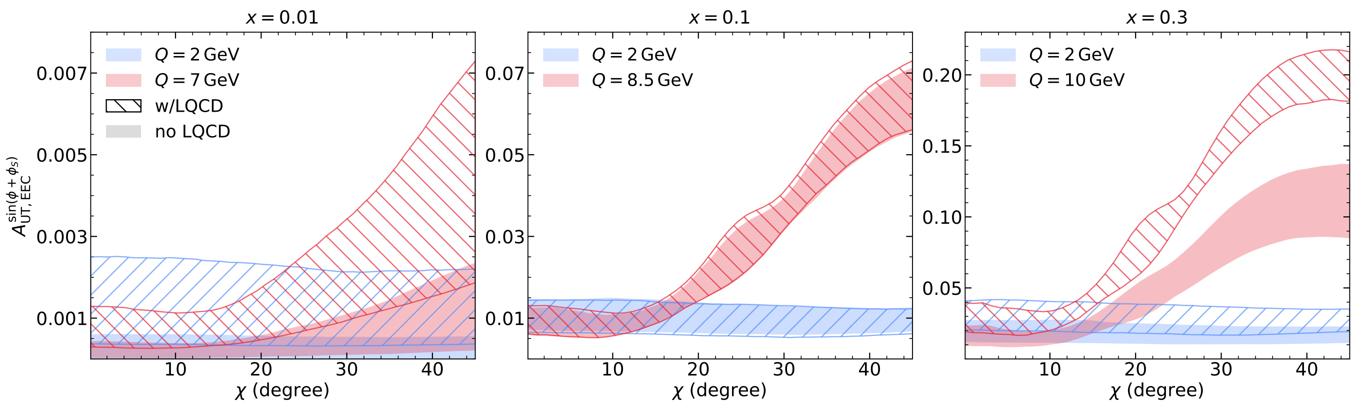

FIG. 1. Predictions for SIDIS $A_{UT,\mathrm{EEC\,SIDIS}}^{\sin(\phi+\phi_S)}$ versus χ at fixed $x = 0.01$, 0.1, 0.3 from left to right. The filled bands shows the central 68% uncertainty band calculated using the JAMDiFF extraction without lattice-QCD priors, while the hatched bands used the extraction with lattice-QCD priors. [Figure edits: Increase size for axis labels, ticks, tick numbers, legend, etc. Need to include in the legend JAMDiFF no LQCD and JAMDiFF w/LQCD.]

theoretically as clean as the one discussed here.]

From the EECs given in Eqs. (9), (10), we can form the following asymmetries that can be measured at the EIC, and BELLE and BESIII, respectively, where we focus on $\pi^+\pi^-$ final states:

$$A_{UT,\mathrm{EEC\,SIDIS}}^{\sin(\phi+\phi_S)} \equiv -\frac{\sum_{q,\bar{q}} e_q^2\, h_1^{q/N}(x)\mathcal{H}^{\pi^+\pi^-/q}(Z)}{\sum_{q,\bar{q}} e_q^2\, f_1^{q/N}(x)\,\mathcal{D}^{\pi^+\pi^-/q}(Z)}\,, \quad (11)$$

$$A_{UT,\mathrm{EEC}\,e^+e^-}^{\cos(\phi+\bar{\phi})} \equiv \frac{\sum_{q,\bar{q}} e_q^2\,\mathcal{H}^{\pi^+\pi^-/q}(Z)\mathcal{H}^{\pi^+\pi^-/\bar{q}}(\bar{Z})}{\sum_{q,\bar{q}} e_q^2\,\mathcal{D}^{\pi^+\pi^-/q}(Z)\mathcal{D}^{\pi^+\pi^-/\bar{q}}(\bar{Z})}\,, \quad (12)$$

where we have removed the depolarization (y-dependent) factors from the definitions. In addition, a measurement is needed of the "standard" EEC in $e^+e^- \to h_1h_2X$ (without the sum over h_1, h_2): [DP: Should we define this more analogous to the previous paper where we divide by σ_t (and also multiply by $(\sin\chi)/2$)?]

$$\begin{aligned}\mathrm{EEC}_{e^+e^-}^{\pi^+\pi^-} &\equiv \int d\tau_1 d\tau_2 d^2\vec{R}_T\,\tau_1\tau_2\frac{d\sigma^{e^+e^-}}{d\tau_1 d\tau_2 d^2\vec{R}_T}\delta(z_\chi - z_{12}) \\ &\overset{\mathrm{DiFF}}{=} \frac{4\pi N_c\alpha_{em}^2}{3Q^2}\sum_{q,\bar{q}} e_q^2\,\mathcal{D}^{\pi^+\pi^-/q}(Z)\,. \quad (13)\end{aligned}$$

One can then simultaneously extract $h_1(x), \mathcal{D}^{\pi^+\pi^-/i}(Z), \mathcal{H}^{\pi^+\pi^-/i}(Z)$ from the data on $A_{UT,\mathrm{EEC\,SIDIS}}^{\sin(\phi+\phi_S)}, A_{UT,\mathrm{EEC}\,e^+e^-}^{\cos(\phi+\bar{\phi})}, \mathrm{EEC}_{e^+e^-}^{\pi}$, analogous to the approach of Ref. [CITE] but with the very helpful simplification of not having to model the M_h dependence of DiFFs.

Modeling of the EEC DiFFs— In order to motivate measurements of Eqs. (9), (10), and (13), we will give numerical predictions for EIC, BELLE, and BESIII kinematics in the next section. We first must model the EEC-DiFFs $\mathcal{D}^{\pi^+\pi^-/i}(Z)$ and $\mathcal{H}^{\pi^+\pi^-/i}(Z)$, and we do so by relating them to the DiFFs $D_1^{\pi^+\pi^-/i}(z, M_h)$ and $H_1^{\sphericalangle\,\pi^+\pi^-/i}(z, M_h)$, respectively, that have been extracted [CITE] from current dihadron measurements in SIDIS, e^+e^- annihilation, and proton-proton collisions [CITE]. However, we emphasize that this approach is only for the purpose of making predictions for this study. In actuality, once experimental data on Eqs. (9), (10), and (13) are available, one would parameterize $\mathcal{D}^{h_1h_2/i}(Z)$ and $\mathcal{H}^{h_1h_2/i}(Z)$ using a functional form like $\sim NQ^2\exp(-Z/\alpha)/(1+Z/\beta)$ mentioned above, bypassing the need to connect them directly to M_h-dependent DiFFs.

As we derive in Supplemental Material, one has the following relation:

$$\begin{aligned}\mathcal{D}^{h_1h_2/i}(Z) =& \frac{Q^2}{32}\int_0^1 d\xi\int_{-1}^1 d\zeta\,\frac{\xi^4(1-\zeta^2)^2}{M_h} \\ &\times D_1^{h_1h_2/i}(\xi,\zeta,M_h)\Big|_{M_h=\widetilde{M}_h}\,, \quad (14)\end{aligned}$$

where $\widetilde{M}_h = \sqrt{Z\xi^2(1-\zeta^2)/4 + M_{h,min}^2}$, with $M_{h,min}^2 = 2M_1^2/(1+\zeta) + 2M_2^2/(1-\zeta)$ the lower bound on M_h^2 [CITE], and $\zeta \equiv (\xi_1 - \xi_2)/\xi$. The exact same expression (14) holds between $\mathcal{H}^{h_1h_2/i}(Z)$ and $H_1^{\sphericalangle\,\pi^+\pi^-/i}(\xi,\zeta,M_h)$. The DiFFs that have been extracted from experiment, though, are only functions of (ξ, M_h) [CITE]. Therefore, we expand the ζ dependence of $D_1^{h_1h_2/i}(\xi,\zeta,M_h)$ and $H_1^{\sphericalangle\,\pi^+\pi^-/i}(\xi,\zeta,M_h)$ in terms of Legendre polynomials up to second order, similar to the partial wave expansion of DiFFs formulated in Ref. [CITE]. Each coefficient of the expansion we assume is proportional to the (ξ, M_h)-dependent DiFF. Specifically,

$$D_1(\xi,\zeta,M_h) \approx D_1(\xi,M_h)\left[\frac{1}{2} + a\zeta + \frac{1}{2}b(3\zeta^2 - 1)\right] \quad (15)$$

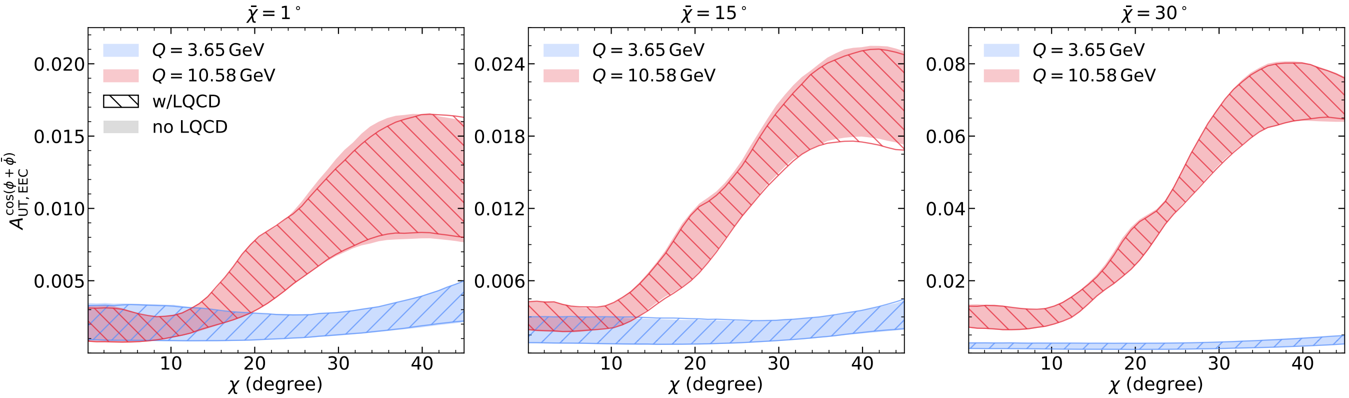

FIG. 2. Predictions for e^+e^- $A^{\cos(\phi+\bar{\phi})}_{UT,\mathrm{EEC}\,e^+e^-}$ versus χ at fixed $\bar{\chi} = 1^\circ, 15^\circ, 30^\circ$ from left to right. Each panel shows predictions for BESIII ($Q = 3.65\,\mathrm{GeV}$) and BELLE ($Q = 10.58\,\mathrm{GeV}$). [Figure edits: Increase size for axis labels, ticks, tick numbers, legend, etc. Need to include in the legend JAMDiFF no LQCD and JAMDiFF w/LQCD.]

$$H_1^{\sphericalangle}(\xi,\zeta,M_h) \approx H_1^{\sphericalangle}(\xi,M_h)\left[\frac{1}{2} + \tilde{a}\zeta + \frac{1}{2}\tilde{b}(3\zeta^2 - 1)\right], \tag{16}$$

where $a, b, \tilde{a}, \tilde{b}$ are unknown constants, and we have dropped the h_1h_2/i superscript on the functions for brevity. Note that this parameterization of the ζ dependence automatically satisfies $\int_{-1}^{1} d\zeta\, D_1(\xi,\zeta,M_h) = D_1(\xi,M_h)$, as must be the case from the number density interpretation of the DiFF operator definitions [CITE]. The positivity bounds on $D_1(\xi,\zeta,M_h)$ and $H_1^{\sphericalangle}(\xi,\zeta,M_h)$, namely, $D_1(\xi,\zeta,M_h) \geq 0$ and $|H_1^{\sphericalangle}(\xi,\zeta,M_h)| \leq D_1(\xi,\zeta,M_h)$ [CITE], also allow one to place constraints on the possible values of a, b, α, β, since the bounds, in particular, must be satisfied for all ζ. Since the positivity bounds also apply to the (ξ, M_h)-dependent DiFFs, the $D_1(\xi,\zeta,M_h) \geq 0$ bound requires $-\frac{1}{2} \leq b \leq 1, |a| \leq \frac{1}{2} + b$, and, when $b > 0$ and $|a| \leq 3b$, $a^2 \leq 3b(1-b)$. The $|H_1^{\sphericalangle}(\xi,\zeta,M_h)| \leq D_1(\xi,\zeta,M_h)$ bound is most naturally satisfied when $\tilde{a} = a$ and $\tilde{b} = b$.

Numerical Predictions— We now give numerical predictions for $\mathrm{EEC}^{\pi}_{e^+e^-}, A^{\sin(\phi+\phi_S)}_{UT,\mathrm{EEC\,SIDIS}}, A^{\cos(\phi+\bar{\phi})}_{UT,\mathrm{EEC}\,e^+e^-}$ in Eqs. (13), (9), (10), utilizing Eqs. (14), (15), (16). We use the JAMDiFF analysis [CITE] as input for $h_1(x), D_1^{\pi^+\pi^-/i}(\xi,M_h)$, $H_1^{\sphericalangle\, h_1h_2/i}(\xi,M_h)$. To generate uncertainty bands, we randomly sample 200 replicas from the JAMDiFF extraction of those functions, and for each of those replicas scan over 100 (a,b) points that are representative of the allowed region for those parameters (see Supplemental Material for details)[Include plot of allowed region and (a,b) points.]. We note that the uncertainty from the choice of values of (a,b) was negligible compared to the (statistical) theoretical uncertainty of the non-perturbative functions [DP: Is this true of $\mathrm{EEC}^{\pi}_{e^+e^-}$?]. We also consider separately the JAMDiFF extraction with and without including priors from lattice QCD on the nucleon tensor charges. The input for $f_1(x)$ is taken from CT18NLO [CITE] and is fixed to the central value.

[DP: What angles should we go out to in each plot?]

In Fig. 1, we present $A^{\sin(\phi+\phi_S)}_{UT,\mathrm{EEC\,SIDIS}}$ vs. χ for various values of Q and fixed values of $x = 0.01, 0.1, 0.3$. We find the asymmetry increases with χ with the maximum shifting towards smaller χ for increasing Q but maintaining the same value. The effect is also largest at larger x values due to $h_1(x)$ peaking in that region, becoming as large as 20% in the JAMDiFF-with-lattice scenario. In Fig. 2, we display $A^{\cos(\phi+\bar{\phi})}_{UT,\mathrm{EEC}\,e^+e^-}$ vs. χ for BELLE and BESIII center-of-mass energies at different fixed values of $\bar{\chi}$. We find the asymmetry is close to 8% at BELLE at the largest $\bar{\chi}$ value but probably not measurable at BESIII.

In Fig. 3, we plot $\mathrm{EEC}^{\pi}_{e^+e^-}$ vs. χ for BELLE ($Q = 10.58\,\mathrm{GeV}$) and BESIII ($Q = 3.65\,\mathrm{GeV}$) kinematics. We find that... We also highlight that there are no resonances in the observable, unlike the cross section measurement $d\sigma/dzdM_h$ from BELLE for $\pi^+\pi^-$ production [CITE], which causes the previously discussed complicated modeling of the M_h-dependence of $D_1^{\pi^+\pi^-/i}(z,M_h)$ in the current DiFF approach to extracting $h_1(x)$.

Overall, our predictions demonstrate the potential of transverse-spin dependent EECs in the near-side (collinear) region to allow for a simplified extraction of $h_1(x)$ compared to the current TMD and dihadron approaches. The phenomenological procedure now becomes analogous to how one can extract $f_1(x), g_1(x)$, and $D_1(z)$ from (single-hadron) DIS, SIDIS, and e^+e^- data in collinear factorization. Our approach can also be extended to proton-proton reactions, e.g., dihadron-in-jet measurements of EECs have very recently become available [CITE]. We leave the development of the theoretical formalism for this observable to future work.

Conclusions and Outlook—

We thank A. Vossen for a helpful discussion on di-

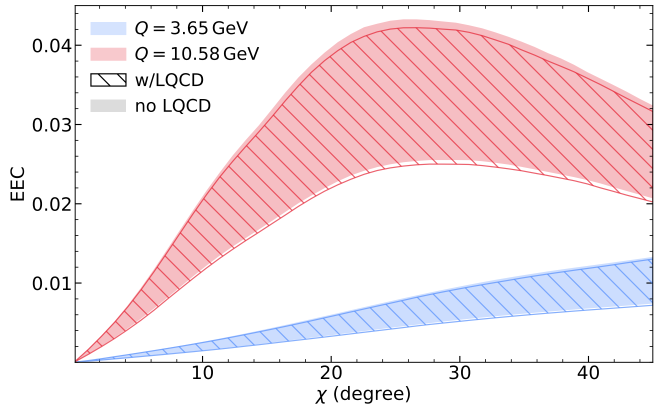

FIG. 3. Predictions for $\mathrm{EEC}^{\pi}_{e^+e^-}$ versus χ at BESIII ($Q = 3.65\,\mathrm{GeV}$, top) and BELLE ($Q = 10.58\,\mathrm{GeV}$, bottom).[Figure edits: Increase size for axis labels, ticks, tick numbers, legend, etc. Need to include in the legend JAMDiFF no LQCD and JAMDiFF w/LQCD. Put together in one plot???, test parameter uncertainty]

hadron experimental measurements. This work was supported by the National Science Foundation under Grants No. PHY-2515057 (Z.K., C.Z.), No. PHY-2110472 and No. PHY-2412792 (A.M.), and No. PHY-2308567 (D.P.).

Data Availability: The data that support the findings of this article are openly available [CITE].

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SUPPLEMENTAL MATERIAL

Unpolarized EEC in Semi-Inclusive DIS: Analytical Result and Numerical Prediction

From Refs. [2, 6] we know that the terms in the cross section $(d\sigma/dxdyd\tau_1 d\tau_2 d^2\vec{R}_T)^{eN\to e'h_1h_2X}$ that only involve dihadron fragmentation will have exactly the same structure as $(d\sigma/dxdyd\tau)^{eN\to e'hX}$ with $D_1^{h/i} \to D_1^{h_1h_2/i}$. However, the replacement $dw/w \to dw/w^2$ is also needed – see, e.g., Eq. (16b) of Ref. [2]. Notably, the partonic cross sections are the same in both cases. Based on the known factorization formula for $(d\sigma/dxdyd\tau)^{eN\to e'hX}$ [7], one can then immediately write down

$$\frac{d\sigma^{eN\to e'h_1h_2X}}{dxdyd\tau_1 d\tau_2 d^2\vec{R}_T} \overset{\text{DiFF}}{=} \frac{2\pi\alpha_{em}^2}{Q^2}\left[\frac{(1+(1-y)^2)}{y}\, 2F_1(x,\tau_1,\tau_2,\vec{R}_T) + \frac{2(1-y)}{y}\, F_L(x,\tau_1,\tau_2,\vec{R}_T)\right] \tag{S1}$$

where

$$\begin{aligned} 2F_1(x,\tau_1,\tau_2,\vec{R}_T) = \sum_{q,\bar{q}} e_q^2 \Bigg\{ & f_1^{q/N}(x)\, D_1^{h_1h_2/q}(\tau_1,\tau_2,\vec{R}_T) \\ & + \frac{\alpha_s}{2\pi}\int_x^1 \frac{d\alpha}{\alpha}\int_\tau^1 \frac{d\beta}{\beta^2}\Big[f_1^{q/N}(\tfrac{x}{\alpha})\, \mathrm{C}^1_{qq}(\alpha,\beta)\, D_1^{h_1h_2/q}(\tfrac{\tau_1}{\beta},\tfrac{\tau_2}{\beta},\vec{R}_T) + f_1^{q/N}(\tfrac{x}{\alpha})\, \mathrm{C}^1_{gq}(\alpha,\beta)\, D_1^{h_1h_2/g}(\tfrac{\tau_1}{\beta},\tfrac{\tau_2}{\beta},\vec{R}_T) \\ & \qquad + f_1^{g/N}(\tfrac{x}{\alpha})\, \mathrm{C}^1_{qg}(\alpha,\beta)\, D_1^{h_1h_2/q}(\tfrac{\tau_1}{\beta},\tfrac{\tau_2}{\beta},\vec{R}_T)\Big]\Bigg\}, \end{aligned} \tag{S2}$$

$$\begin{aligned} F_L(x,\tau_1,\tau_2,\vec{R}_T) = \frac{\alpha_s}{2\pi}\sum_{q,\bar{q}} e_q^2 \int_x^1 \frac{d\alpha}{\alpha}\int_\tau^1 \frac{d\beta}{\beta^2}\Big[& f_1^{q/N}(\tfrac{x}{\alpha})\, \mathrm{C}^L_{qq}(\alpha,\beta)\, D_1^{h_1h_2/q}(\tfrac{\tau_1}{\beta},\tfrac{\tau_2}{\beta},\vec{R}_T) \\ & + f_1^{q/N}(\tfrac{x}{\alpha})\, \mathrm{C}^L_{gq}(\alpha,\beta)\, D_1^{h_1h_2/g}(\tfrac{\tau_1}{\beta},\tfrac{\tau_2}{\beta},\vec{R}_T) + f_1^{g/N}(\tfrac{x}{\alpha})\, \mathrm{C}^L_{qg}(\alpha,\beta)\, D_1^{h_1h_2/q}(\tfrac{\tau_1}{\beta},\tfrac{\tau_2}{\beta},\vec{R}_T)\Big]\Bigg\}, \end{aligned} \tag{S3}$$

with $\mathrm{C}^{1,L}_{ij}$ being the NLO coefficient functions that can be found in Appendix C of Ref. [7]. From here we can calculate a "standard" EEC for SIDIS, defined as [DP: Should we define this more analogous to the previous paper where we divide by σ_t (and also multiply by $(\sin\chi)/2$)?]

$$\begin{aligned} \mathrm{EEC}_{\mathrm{SIDIS}} &\equiv \sum_{h_1,h_2}\int d\tau_1 d\tau_2 d^2\vec{R}_T\, \tau_1\tau_2\, \frac{d\sigma^{eN\to e'h_1h_2X}}{dxdyd\tau_1 d\tau_2 d^2\vec{R}_T}\,\delta(z_\chi - z_{12}) \\ &\overset{\text{DiFF}}{=} \left(\frac{2\pi\alpha_{em}^2}{Q^2}\sum_{q,\bar{q}} e_q^2\right)\left[\frac{(1+(1-y)^2)}{y}\, 2\mathcal{F}_1(x,Z) + \frac{2(1-y)}{y}\, \mathcal{F}_L(x,Z)\right], \end{aligned} \tag{S4}$$

where

$$\begin{aligned} 2\mathcal{F}_1(x,Z) = \Bigg\{ f_1^{q/N}(x)\,\mathcal{D}^q(Z) + \frac{\alpha_s}{2\pi}\int_x^1\frac{d\alpha}{\alpha}\int_\tau^1 d\beta\,\beta^2\Big[& f_1^{q/N}(\tfrac{x}{\alpha})\,\mathrm{C}^1_{qq}(\alpha,\beta)\,\mathcal{D}^q(\beta^2 Z) + f_1^{q/N}(\tfrac{x}{\alpha})\,\mathrm{C}^1_{gq}(\alpha,\beta)\,\mathcal{D}^g(\beta^2 Z) \\ & + f_1^{g/N}(\tfrac{x}{\alpha})\,\mathrm{C}^1_{qg}(\alpha,\beta)\,\mathcal{D}^q(\beta^2 Z)\Big]\Bigg\}, \end{aligned} \tag{S5}$$

$$\begin{aligned} \mathcal{F}_L(x,Z) = \frac{\alpha_s}{2\pi}\int_x^1\frac{d\alpha}{\alpha}\int_\tau^1 d\beta\,\beta^2\Big[& f_1^{q/N}(\tfrac{x}{\alpha})\,\mathrm{C}^L_{qq}(\alpha,\beta)\,\mathcal{D}^q(\beta^2 Z) + f_1^{q/N}(\tfrac{x}{\alpha})\,\mathrm{C}^L_{gq}(\alpha,\beta)\,\mathcal{D}^g(\beta^2 Z) \\ & + f_1^{g/N}(\tfrac{x}{\alpha})\,\mathrm{C}^L_{qg}(\alpha,\beta)\mathcal{D}^q(\beta^2 Z)\Big]\Bigg\}, \end{aligned} \tag{S6}$$

with $Z \equiv z_\chi Q^2$ and $\mathcal{D}^q(Z), \mathcal{D}^g(Z)$ the EEC-DiFFs extracted in Ref. [3], and we have assumed $\mathcal{D}^q(Z)$ is flavor independent. Therefore, we are able to make a LO prediction for $\mathrm{EEC}_{\mathrm{SIDIS}}$ using those functions from Ref. [3] as input.

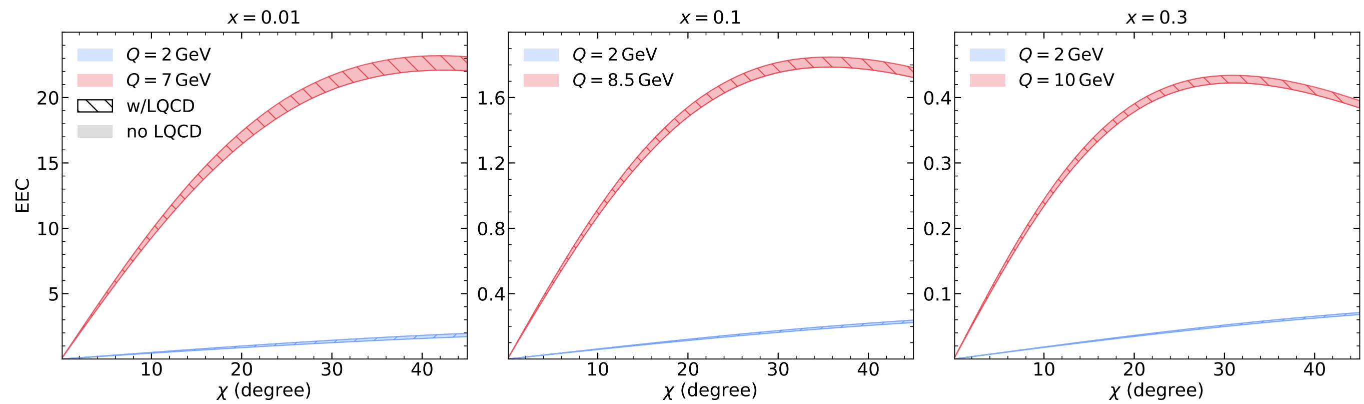

FIG. S4. Leading-order predictions for the unpolarized SIDIS EEC versus χ at fixed $x = 0.01,\, 0.1,\, 0.3$ from left to right. The filled bands show the central 68% uncertainty band calculated using the JAMDiFF extraction without lattice-QCD priors, while the hatched bands use the extraction with lattice-QCD priors.

More Details on the Modeling of the EEC DiFFs $\mathcal{D}^{\pi^+\pi^-/i}(Z)$ and $\mathcal{H}^{\pi^+\pi^-/i}(Z)$

Starting from Eq. (4) in the main text, we can use the relation [1]

$$D_1^{h_1h_2/i}(\xi_1,\xi_2,\vec{R}_T) = \frac{4}{\pi\xi M_h(1-\zeta^2)}\, D_1^{h_1h_2/i}(\xi,\zeta,M_h)\,, \tag{S7}$$

along with the variable transformation $d\xi_1 d\xi_2 d^2\vec{R}_T = \frac{\xi(1-\zeta^2)}{16} d\xi d\zeta d\phi_{R_T} dM_h^2$, and the fact that

$$\frac{R_T^2}{Q^2}\frac{\xi^2}{\xi_1^2\xi_2^2} = \frac{4}{Q^2\xi^2(1-\zeta^2)}(M_h^2 - M_{h,min}^2)\,, \tag{S8}$$

where is $M_{h,min}^2 = 2M_1^2/(1+\zeta) + 2M_2^2/(1-\zeta)$, $\xi \equiv \xi_1+\xi_2$, and $\zeta \equiv (\xi_1-\xi_2)/\xi$, to write

$$\mathcal{D}^{h_1h_2/i}(Z) = \frac{Q^2}{32}\int_0^1 d\xi \int_{-1}^1 d\zeta\, \frac{\xi^4(1-\zeta^2)^2}{M_h} D_1^{h_1h_2/i}(\xi,\zeta,M_h)\Big|_{M_h=\widetilde{M}_h}\,, \tag{S9}$$

where $\widetilde{M}_h = \sqrt{Z\xi^2(1-\zeta^2)/4 + M_{h,min}^2}$. This is Eq. (14) in the main text. An anologous derivation allows one to write down the exact same expression for $\mathcal{H}^{h_1h_2/i}(Z)$ and $H_1^{\sphericalangle\,\pi^+\pi^-/i}(\xi,\zeta,M_h)$.

Now we turn to the Legendre expansions in Eqs. (15), (16) and provide more details on the region of (a,b) explored in generating our numerical predictions. Since $D_1(\xi,M_h) \geq 0$, we must have $f(\zeta) = \frac{1}{2} + a\zeta + \frac{1}{2}b(3\zeta^2-1) \geq 0$ $\forall\zeta \in [-1,1]$ in order for $D_1(\xi,\zeta,M_h) \geq 0$ to also be satisfied. The critical point of $f(\zeta)$ occurs at $\zeta_0 = -\frac{a}{3b}$. If $b>0$, then $f(\zeta)$ is a concave up parabola and ζ_0 gives a minimum. If $|a| \leq 3b$ then $\zeta_0 \in [-1,1]$, and it must be required that $f(\zeta_0) \geq 0$, which implies $a^2 \leq 3b(1-b)$. Otherwise, it is the value of $f(\zeta)$ at the endpoints $\zeta=-1$ and $\zeta=1$ that must be ≥ 0, giving us the condition $|a| \leq b + \frac{1}{2}$, which also implies $b \geq -\frac{1}{2}$. At $\zeta = 0$, $f(\zeta) \geq 0$ also restricts $b \leq 1$, and, therefore, $-\frac{1}{2} \leq b \leq 1$. If $b < 0$, then $f(\zeta)$ is a concave down parabola. Even if $\zeta_0 \in [-1,1]$, $f(\zeta_0)$ is a maximum, so what matters is only that $f(\zeta=-1)$ and $f(\zeta=1)$ are both ≥ 0, and we revert back to the condition $|a| \leq b+\frac{1}{2}$.